

① Derivation 1:

Show for single particle; constant mass:

i) $\frac{\partial T}{\partial t} = \vec{F} \cdot \vec{v}$ and if mass varies w time:

ii) $\frac{d(mT)}{dt} = \vec{F} \cdot \vec{p}$

for a single particle: $\vec{p} = m\vec{v}$, $\vec{F} = \frac{d\vec{p}}{dt}$

Kinetic energy is $T = \frac{1}{2}mv^2$

↳ differentiate: $\frac{\partial T}{\partial t} = \frac{1}{2}m \cdot \frac{d}{dt}(\vec{v} \cdot \vec{v}) = m\vec{v} \cdot \vec{v}$

$$\boxed{\frac{\partial T}{\partial t} = m \cdot v \cdot \vec{v} = \vec{F} \cdot \vec{v}}$$

rate of change of KE
is
instantaneous power delivered
by the force.

Variable mass:

KE in terms of momentum: $T = \frac{p^2}{2m}$; $\vec{F} = \dot{\vec{p}}$; $p = mv$

$$(m)T = (m)\frac{p^2}{2m} = \frac{1}{2}p^2 = \frac{1}{2}\vec{p} \cdot \vec{p}$$

$$\frac{d(mT)}{dt} = \vec{p} \cdot \frac{d\vec{p}}{dt}$$

and since $\vec{F} = \frac{d\vec{p}}{dt}$

$$\downarrow \boxed{\frac{d(mT)}{dt} = \vec{F} \cdot \vec{p}}$$

Derivation (2): Starting w/ $\dot{r} = \sqrt{\frac{2}{m} (E - V(r) - \frac{\ell^2}{2mr^2})}$

i) Introduce quasi pot. $V'(r)$ & use COE_{Total} no discs

(a) unbounded
 (b) bounded
 (c) circular trajectories

for $1/r^2$ forces

$[V(r) = -\frac{k}{r}, k > 0]$

Let: $\frac{1}{2} m \dot{r}^2 + V'(r) = E$ and $V'(r) = V(r) + \frac{\ell^2}{2mr^2}$

for $V(r) = -\frac{k}{r}$ we have: $V'(r) = \frac{\ell^2}{2mr^2} - \frac{k}{r} \dots (1)$

So radial motion is allowed when: $E \geq V'(r)$

and $V'(r) \rightarrow \infty$ ($r \rightarrow 0$)

bc of centrifugal term $V'(r) \rightarrow 0$ ($r \rightarrow \infty$)

which gives us a minimum between these limits

2i: (a) unbounded:

- at large r , $V'(r) \rightarrow 0^-$

therefore particle can reach $r \rightarrow \infty$

for $E = 0$; particle reaches infinity w/ zero limiting speed.

$E = 0$; parabolic orbit; unbounded

2i: (b) Let E_{min} be minimum of $V'(r)$

for $E_{min} < E < 0$ there is a horizontal entry line

that intersects $V'(r)$ at (2) radii $[E = V'(r)]$

$$r_{min} \leq r \leq r_{max}$$

hence particle cannot escape to infinity:

$E_{min} < 0 < E$: bounded elliptical orbit

(c) circular: $r = r_0 = \text{constant}$; $\dot{r} = 0$

↳ Energy must lie exactly at quasipotential

$$\left. \frac{\partial V'(r)}{\partial r} \right|_0 = 0$$

and $E = V'(r_0) = E_{min}$

↳ Since stationary point will be a minimum

$$\left. \frac{\partial^2 V'(r)}{\partial r^2} \right|_{r_0} > 0$$

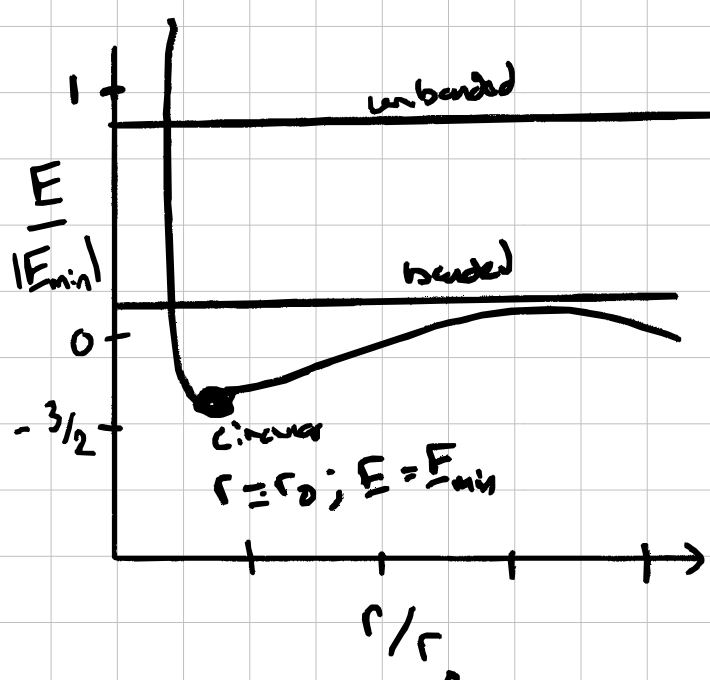
circular orbit is
Stable.

$E > 0 \rightarrow$ unbounded hyperbolic

$E = 0 \rightarrow$ unbounded parabolic

$E_{min} < E < 0 \rightarrow$ bounded elliptical

$E = E_{min} \rightarrow$ circular



2ii) find r_0 and E in circular case:

for circular: $\frac{\partial V'(r)}{\partial r} = 0$] Since $V'(r) = \frac{\ell^2}{2mr^2} - \left(-\frac{k}{r}\right)$

$$\frac{\partial V'(r)}{\partial r} = -\frac{\ell^2}{mr^3} + \frac{k}{r^2} \dots (2)$$

$$\text{at } r=r_0 \quad -\frac{\ell^2}{mr_0^3} + \frac{k}{r_0^2} = 0$$

$$\hookrightarrow \boxed{r_0 = \frac{\ell^2}{mk}}$$

$$\hookrightarrow \text{check min: } \frac{\partial^2 V'(r)}{\partial r^2} = \frac{3\ell^2}{mr^4} - \frac{2k}{r^3} \quad \text{and } \frac{\ell^2}{m} = kr_0$$

$$\left. \frac{\partial^2 V'(r)}{\partial r^2} \right|_{r_0} = \frac{3k}{r_0^3} - \frac{2k}{r_0^3} = \frac{k}{r_0^3} > 0$$

$$\text{for total Energy: } E = V'(r_0) \text{ thus: } E = \frac{\ell^2}{2mr_0^2} - \frac{k}{r_0}$$

$$\text{recall again } \frac{\ell^2}{m} = kr_0 \text{ then } \frac{\ell^2}{2mr_0^2} = \frac{k}{2r_0}$$

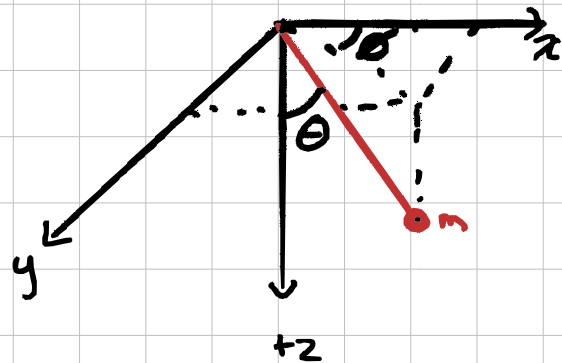
$$\text{so: } E = \frac{k}{2r_0} - \frac{k}{r_0} = -\frac{k}{2r_0} \quad \text{and since } r_0 = \frac{\ell^2}{mk}$$

$$\boxed{E = -\frac{k}{2r_0} = -\frac{mk^2}{2\ell^2}}$$

$$\boxed{r_0 = \frac{\ell^2}{mk}}$$

Problem ①: Spherical Pendulum

→ write $\mathcal{L}$ in spherical



Let:

θ measured from downward $+z$, ϕ is azimuth

→ rod is fixed $\therefore r = l$

so:

$$x = l \sin \theta \cos \phi, \quad y = l \sin \theta \sin \phi, \quad z = l \cos \theta$$

$$\dot{x} = l (\dot{\theta} \cos \theta \cos \phi - \dot{\phi} \sin \theta \sin \phi)$$

$$\dot{y} = l (\dot{\theta} \cos \theta \sin \phi + \dot{\phi} \sin \theta \cos \phi)$$

$$\dot{z} = -l \dot{\theta} \sin \theta$$

} will get nice cancellations

then

$$v^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2$$

$$= l^2 \left[\dot{\theta}^2 \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + \dot{\phi}^2 \sin^2 \theta (\sin^2 \phi + \cos^2 \phi) - 2 \dot{\theta} \dot{\phi} \cos \theta \sin \theta \cos \phi \sin \phi + 2 \dot{\theta} \dot{\phi} \cos \theta \sin \theta \cos \phi + \dot{\theta}^2 \sin^2 \theta \right]$$

→ mixed terms cancel and $\sin^2 \alpha + \cos^2 \alpha = 1$ so

$$v^2 = l^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)$$

$$\therefore T = \frac{1}{2} m l^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) \quad \dots (1)$$

$+z$ is defined in pic as down so:

$$V = -mgz = -mg l \cos \theta \quad \dots (2)$$

$$\mathcal{L} = T - V = \frac{m l^2}{2} (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + (mg l \cos \theta)$$

Problem does have same time: will probe $E-l$

$$\mathcal{L} = \frac{m\ell^2}{2} (\dot{\theta}^2 + \sin^2 \theta \cdot \dot{\phi}^2) + mg\ell \cos \theta$$

for θ

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m\ell^2 \dot{\theta} \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = m\ell^2 \ddot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = m\ell^2 \sin \theta \cos \theta \dot{\phi}^2 - mg\ell \sin \theta$$

$$\hookrightarrow \ddot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 + \frac{g}{\ell} \sin \theta = 0 \quad \dots (a)$$

for ϕ

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0; \quad \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = m\ell^2 \sin^2 \theta \cdot \dot{\phi}$$

$$\Rightarrow \frac{d}{dt} (m\ell^2 \sin^2 \theta \dot{\phi}) = 0$$

$$\hookrightarrow \sin^2 \theta \cdot \ddot{\phi} + 2 \sin \theta \cos \theta \cdot \dot{\theta} \dot{\phi} = 0 \quad \dots (b)$$

here ϕ is cyclic bc $\mathcal{L}$ is independent of ϕ hence

~~conjugate~~ $P_{\phi} = m\ell^2 \sin^2 \theta \dot{\phi}$; angular momentum about vertical axis is conserved.